

Sign Reversal and the Exact Zero Set for the Covariance of Competing Bulk and Surface Stopping in the Three-Ball

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Abstract

For reflected Brownian motion in the unit three-ball, started from the volume-uniform distribution, consider an independent exponential bulk clock of rate p and an exponential boundary-local-time threshold of rate q . Let T be the first of the two stopping times and let L be the local time accumulated at T . A closed formula in recent work was accompanied by positive values over the parameter range explored there. We show that positivity does not extend to the whole quadrant. An independent moment derivation gives

$$\text{Cov}(T, L) = \frac{3\{\xi_1(p) + q\xi_2(p)\}}{4p^3\{\sqrt{p}\cosh(\sqrt{p}) - \sinh(\sqrt{p}) + q\sinh(\sqrt{p})\}^2}.$$

Here $\xi_1(p) > 0$ and $\xi_2(p) < 0$ for every $p > 0$. Consequently the complete zero set is the graph $q = -\xi_1(p)/\xi_2(p)$, with positive covariance below it and negative covariance above it. In particular, $(p, q) = (1, 200)$ is an exact counterexample to universal positivity.

1 Model and result

Let

$$\mathcal{B} = \{z \in \mathbb{R}^3 : |z| < 1\}$$

and let $(X_t)_{t \geq 0}$ be reflected Brownian motion in $\overline{\mathcal{B}}$ with interior generator Δ . Its initial point is independent and volume-uniform in $\mathcal{B}$. We use the boundary-local-time normalization of Ye [1]: with ℓ_t denoting local time, the corresponding Robin parameter is the stopping rate appearing below. Let

$$\delta \sim \text{Exp}(p), \quad \widehat{\ell} \sim \text{Exp}(q), \quad p, q > 0,$$

independently of each other, of X_0 , and of the reflected path, and set

$$\tau = \inf\{t > 0 : \ell_t > \widehat{\ell}\}, \quad T = \min\{\delta, \tau\}, \quad L = \ell_T.$$

Continuity of local time and the continuous thresholds give $L = \min\{\ell_\delta, \widehat{\ell}\}$ almost surely.

For $x = \sqrt{p}$, define

$$S = \sinh x, \quad C = \cosh x, \quad A = xC - S, \quad B = A + qS,$$

and

$$\xi_1(x^2) = 4x^2 A^2, \tag{1}$$

$$\xi_2(x^2) = 2x^2(x^2 + 3) - x(x^2 + 12)\sinh(2x) + 6(x^2 + 1)\cosh(2x) - 6. \tag{2}$$

These are the two affine-numerator functions in Appendix F of Ye [1].

Theorem 1. *For every $p, q > 0$,*

$$\text{Cov}(T, L) = \frac{3\{\xi_1(p) + q\xi_2(p)\}}{4x^6 B^2}, \quad x = \sqrt{p}. \quad (3)$$

Moreover, $\xi_1(p) > 0$ and $\xi_2(p) < 0$ for every $p > 0$. Thus the zero set of $\text{Cov}(T, L)$ in $(0, \infty)^2$ is exactly

$$q = q_0(p) := -\frac{\xi_1(p)}{\xi_2(p)}, \quad p > 0. \quad (4)$$

The covariance is positive for $0 < q < q_0(p)$, zero for $q = q_0(p)$, and negative for $q > q_0(p)$. In particular,

$$\text{Cov}_{p=1, q=200}(T, L) < 0.$$

The theorem disproves the assertion that the covariance is positive throughout the positive parameter quadrant. The variances of both T and L are finite and strictly positive there, so the same classification holds for the sign of their Pearson correlation.

2 Splitting probability and moments

Write

$$\phi(p, q) = \mathbb{P}(\tau < \delta).$$

We first recover this splitting probability directly from the model. If $u(r)$ is the probability transform before averaging the initial point, the Feynman–Kac/Robin problem under the stated local-time normalization is

$$(\Delta - p)u = 0 \quad (0 < r < 1), \quad u_r(1) = q\{1 - u(1)\}.$$

The regular radial solution has the form $u(r) = c \sinh(xr)/r$. Since $u_r(1) = cA$, the boundary condition gives $c = q/B$. Volume averaging therefore yields

$$\phi = 3 \int_0^1 u(r) r^2 dr = \frac{3qA}{x^2 B}. \quad (5)$$

Notice that $A(0) = 0$ and $A'(x) = x \sinh x > 0$, so $A > 0$, and hence $B > 0$, whenever $x, q > 0$.

We next derive the moments without assuming the Appendix-F covariance formula. Conditional on the reflected path,

$$\mathbb{P}(T > t \mid X) = e^{-pt - q\ell_t}, \quad \phi = \mathbb{E}(1 - e^{-q\ell_\delta}).$$

Tonelli's theorem and differentiation under these nonnegative, exponentially dominated integrals give

$$\mathbb{E}T = \frac{1 - \phi}{p}, \quad \mathbb{E}L = \frac{\phi}{q}, \quad (6)$$

$$\mathbb{E}T^2 = \frac{2(1 - \phi)}{p^2} + \frac{2\phi_p}{p}, \quad \mathbb{E}L^2 = \frac{2\phi}{q^2} - \frac{2\phi_q}{q}. \quad (7)$$

For the mixed moment, let $H_l = \inf\{t : \ell_t > l\}$. Splitting the joint-survival integral for $\mathbb{E}(TL)$ into $t < H_l$ and $t > H_l$ gives

$$\begin{aligned} \mathbb{E}(TL) &= \int_0^\infty \mathbb{E}[H_l e^{-pH_l - ql}] dl + \int_0^\infty \mathbb{E}[\ell_t e^{-pt - q\ell_t}] dt \\ &= -\frac{\phi_p}{q} + \frac{\phi_q}{p}. \end{aligned}$$

Combining this identity with (6), we obtain the useful formula

$$\text{Cov}(T, L) = \frac{-p\phi_p + q\phi_q + (\phi - 1)\phi}{pq}. \quad (8)$$

Direct differentiation of (5) gives

$$\phi_q = \frac{3A^2}{x^2B^2}, \quad \phi_p = \frac{3q\{xq(SC - x) - 2AB\}}{2x^4B^2}. \quad (9)$$

Substitution of (5) and (9) into (8), followed by the identity $AC - xS^2 = x - SC$, gives

$$\text{Cov}(T, L) = \frac{3}{4x^6B^2} \left[4x^2A^2 + q\{2x^3(x - SC) + 12A^2\} \right]. \quad (10)$$

Double-angle identities show that the expression multiplying q in (10) is precisely $\xi_2(x^2)$ from (2), while the other term is $\xi_1(x^2)$. This proves (3); importantly, its prefactor is strictly positive.

3 Global sign and zero set

It remains to determine the sign of the two numerator functions. The first is immediate: $x > 0$ and $A > 0$, so (1) gives $\xi_1(x^2) > 0$.

For the second, expand the entire functions in (2). The coefficients through degree eight cancel. For $n \geq 3$, before the zero coefficients at $n = 3, 4$ are removed, the coefficient of x^{2n} is

$$\frac{2^{2n-3}}{(2n)!} \{-m(m-1)(m-2) - 48m + 12m(m-1) + 48\}, \quad m = 2n.$$

The polynomial in braces factors exactly as

$$-(m-1)(m-6)(m-8).$$

Consequently, for every $x > 0$,

$$\xi_2(x^2) = - \sum_{n=5}^{\infty} \frac{2^{2n-3}(2n-1)(2n-6)(2n-8)}{(2n)!} x^{2n} < 0. \quad (11)$$

All summands after the minus sign are positive. Equations (3) and (11) now show that, for fixed $p > 0$, the covariance numerator is a strictly decreasing affine function of q , positive at $q = 0$, and has the unique positive root (4). Since the prefactor in (3) is positive, this proves the complete sign and zero-set classification in Theorem 1.

4 An exact parameter counterexample

The global classification already gives counterexamples. We now certify one using rational inequalities only. Take $p = 1$, $q = 200$. Then $x = 1$, $A = e^{-1}$, and, with $y = e^2$, exact algebra gives

$$2y\{\xi_1(1) + 200\xi_2(1)\} = 8 - 200(y^2 - 4y - 25). \quad (12)$$

The positive-term exponential series yields

$$e^2 > \sum_{n=0}^{11} \frac{2^n}{n!} = \frac{164591}{22275} = \frac{7389}{1000} + \frac{41}{891000}.$$

The function $f(y) = y^2 - 4y - 25$ is strictly increasing for $y > 2$, and

$$f\left(\frac{7389}{1000}\right) = \frac{41321}{10^6} > \frac{1}{25}.$$

It follows that the right-hand side of (12) is strictly negative. Since $2y > 0$, the numerator in (3) is negative, and therefore $\text{Cov}_{1,200}(T, L) < 0$ exactly.

5 Variance and literature scope

The correlation has the same sign as the covariance. Indeed, $0 \leq T \leq \delta$ and $0 \leq L \leq \widehat{\ell}$, so both second moments are finite. For every $t > 0$, $\mathbb{P}(T < t) \geq \mathbb{P}(\delta < t) > 0$, while

$$\mathbb{P}(T > t) = \mathbb{E}(e^{-pt-q\ell_t}) > 0,$$

because finite-time reflected-Brownian local time is almost surely finite. Thus T is nonconstant. If $H_{\partial\mathcal{B}}$ is the first boundary-hitting time, the volume-uniform starting point is in the interior almost surely and

$$\mathbb{P}(\delta < H_{\partial\mathcal{B}}) = \mathbb{E}(1 - e^{-pH_{\partial\mathcal{B}}}) > 0.$$

On this event $L = 0$. On the other hand, $\mathbb{E}L = 3A/(x^2B) > 0$ by (6) and (5), so $L > 0$ with positive probability. Hence both variances are strictly positive throughout the open quadrant.

Ye [1] derives the distributions, joint transform, and mixed moments for the two competing mechanisms and gives explicit three-ball formulae; the present result addresses the full-quadrant sign question suggested by the positive values displayed over the range explored there. We make no broader claim of bibliographic priority. As a methodological comparison only, Zeng [2] combines analytic probability bounds with independently certified exact finite comparisons in an unrelated problem about permutation cycle lengths. That paper supplies no input about reflected diffusion, local time, or the covariance formula here; the limited parallel is the separation of analytic proof from exact computational checks.

Computational and AI assistance disclosure

SymPy 1.13.3 was used to check the differentiations, algebraic identities, formal-series coefficients, and rational arithmetic appearing above. A separate Python standard-library check independently verified the formal series through degree 80, the displayed general coefficient factorization, and the rational counterexample certificate. OpenAI Codex assisted with organization, drafting, and algebraic cross-checking. These tools were not treated as proofs: the sign argument and exact certificate are given explicitly in the text.

References

- [1] Yilin Ye. Diffusion under competing bulk and surface stopping mechanisms, 2026. arXiv:2609.05247v1, submitted 4 September 2026.
- [2] Zijian Zeng. The distinct-cycle-length probability is strictly decreasing after 30. SSRN, 2026.